

## EBS vs S3

**Tasks-1.** Create a table comparing Amazon EBS and Amazon S3, listing at least 4 key differences (such as data type, access method, use case, and scalability) and give one real-world app example where each would be the better choice.

| Feature       | Amazon EBS                                           | Amazon S3                                                    |
|---------------|------------------------------------------------------|--------------------------------------------------------------|
| Data type     | Block storage (disk volumes)                         | Object storage (files and objects)                           |
| Access method | Attached to an EC2 instance                          | Accessed through APIs, SDKs, or the AWS Console              |
| Main use case | Operating systems, databases, and application disks  | Images, videos, documents, backups, and static website files |
| Scalability   | Increase volume size or provision additional volumes | Scales to store very large amounts of data                   |
| Availability  | Typically tied to a specific Availability Zone       | Designed for high durability across multiple facilities      |
| Performance   | Low-latency block-level access                       | Object-based access over a network                           |
| Example       | A database storing live order records                | A bucket storing restaurant menu images                      |

Real-world examples:

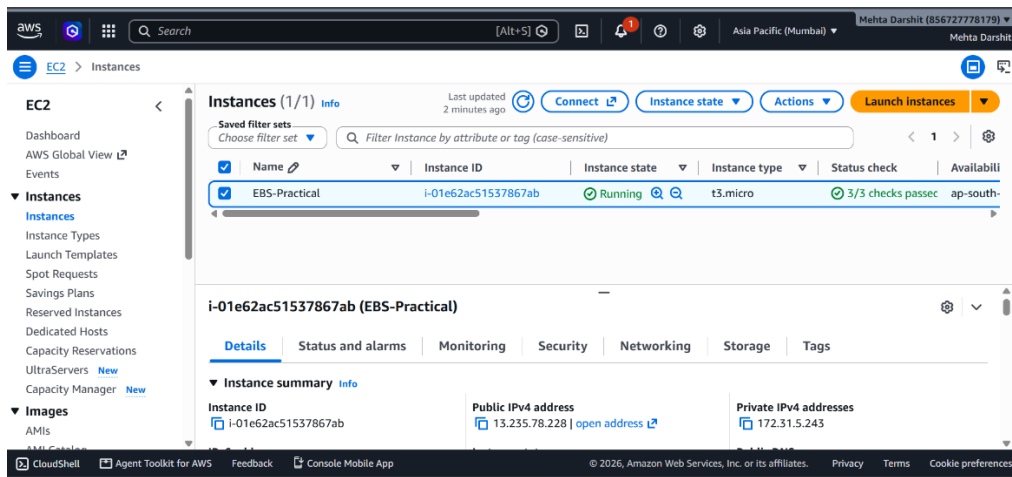
- EBS: A Flipkart order-tracking application could use EBS-backed storage for a database running on EC2, where frequent reads and writes are required.
- S3: A Zomato-style application could use S3 to store restaurant photos, menu images, and other uploaded media.

**Tasks-2.** Launch a new EC2 instance in the AWS Free Tier and attach a new EBS volume of at least 2 GB to it using the AWS Management Console. After launching the instance, use the 'Volumes' section in EC2 to create and attach the volume.

Step 1 — Launch an EC2 instance

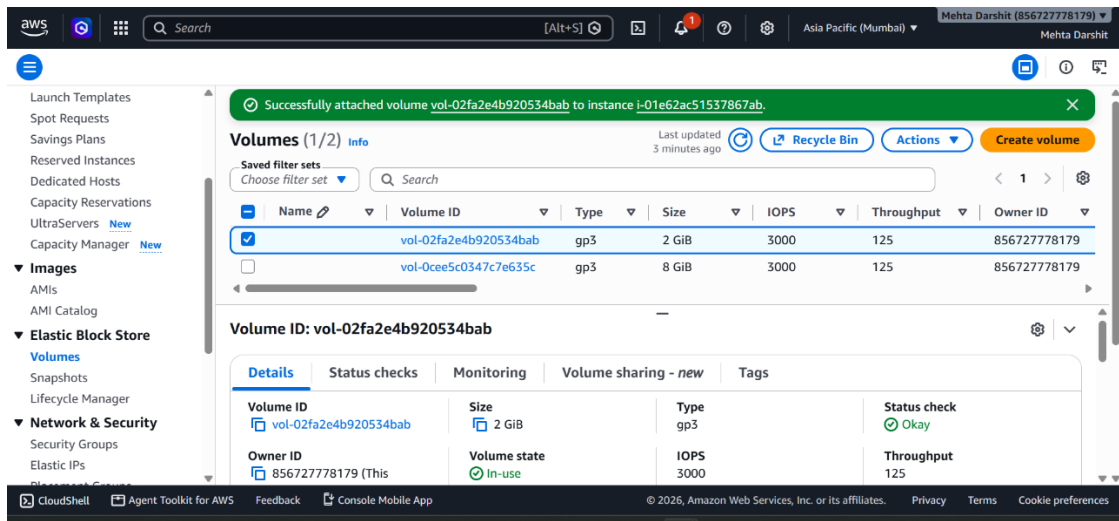
1. Open the EC2 Console

2. Select Launch instance.
3. Enter a name, such as EBS-Practical.
4. Choose an Amazon Linux AMI and a Free Tier-eligible instance type, if available for your account.
5. Select or create a key pair.
6. Allow SSH (port 22) from your IP address.
7. Launch the instance.



## Step 2 — Create and attach a volume

1. In EC2, go to Elastic Block Store → Volumes.
2. Click Create volume.
3. Choose gp3 (General Purpose SSD).
4. Set the size to 2 GiB or more.
5. Select the same Availability Zone as your EC2 instance.
6. Click Create volume.
7. Select the new volume → Actions → Attach volume.
8. Choose your instance and attach it.



**Tasks-3.** Log in to your EC2 instance using SSH and format the attached EBS volume, then mount it to a new directory named `/mnt/mydata`. Use `lsblk` to find your volume, `mkfs` to format, and mount to attach it to the directory.

Connect to your EC2 instance using SSH. For Amazon Linux, you can use the following commands.

### 1. Find the new volume

`lsblk`

Example output:

```
NAME      SIZE TYPE MOUNTPOINTS
nvme0n1   8G  disk
└─nvme0n1p1 8G  part /
nvme1n1   2G  disk
```

In this example, `/dev/nvme1n1` is the new volume. Your device name may differ.

Important: Identify the new, empty 2 GiB volume carefully. Never format your root disk or a volume containing data you need.

## 2. Format the volume

Replace `/dev/nvme1n1` with your actual device name.

```
sudo mkfs -t ext4 /dev/nvme1n1
```

This creates an ext4 filesystem on the new volume.

## 3. Create the mount directory and mount it

```
sudo mkdir -p /mnt/mydata
```

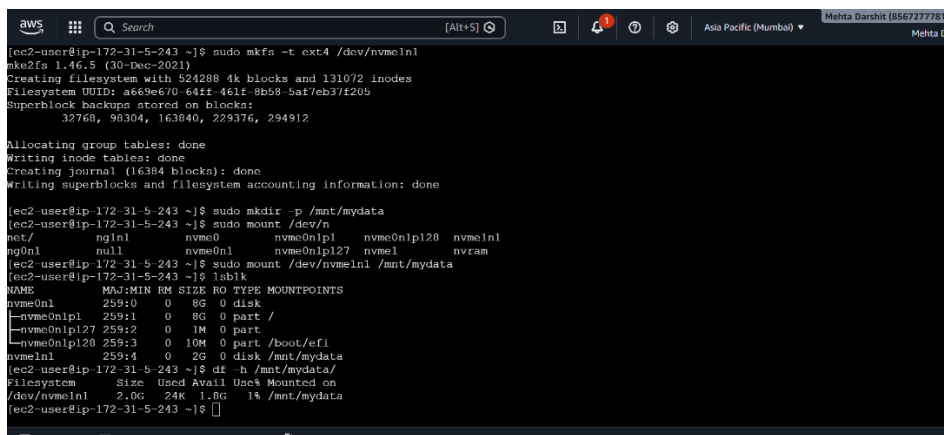
```
sudo mount /dev/nvme1n1 /mnt/mydata
```

Verify the mount:

```
lsblk
```

```
df -h /mnt/mydata
```

You should see the volume mounted at `/mnt/mydata`.



```
aws
[ec2-user@ip-172-31-5-243 ~]$ sudo mkfs -t ext4 /dev/nvme1n1
mkfs 1.46.5 (30-Dec-2021)
Creating filesystem with 924288 4k blocks and 131072 inodes
Filesystem UUID: a669e670-64ff-4611-8b58-baf7eb37f205
Superblock backups stored on blocks:
32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-172-31-5-243 ~]$ sudo mkdir -p /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ sudo mount /dev/nvme1n1 /mnt/mydata
net/      ng1n1     nvme0     nvme0n1p1  nvme0n1p28  nvme1n1
ng0n1     null      nvme0n1   nvme0n1p127  nvme1       nvram
[ec2-user@ip-172-31-5-243 ~]$ sudo mount /dev/nvme1n1 /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ lsblk
NAME        MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0 8G  0 disk 
├─nvme0n1p1 259:1    0 8G  0 part /
├─nvme0n1p127 259:2    0 1M  0 part 
└─nvme0n1p128 259:3    0 10M  0 part /boot/efi
nvme1n1     259:4    0 2G  0 disk /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ df -h /mnt/mydata/
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme1n1    2.0G  24K  1.9G   1% /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$
```

**Tasks-4.** Simulate a Zomato-style image upload: create a text file named `zomato_menu.txt` on your EC2 instance, copy it to your mounted EBS volume, and verify it is accessible from the new mount point.

## 1. Create the text file

```
echo "Zomato Menu: Pizza, Burger, Biryani, Pasta" > ~/zomato_menu.txt
```

## 2. Copy it to the EBS volume

```
sudo cp ~/zomato_menu.txt /mnt/mydata/
```

### 3. Verify the file

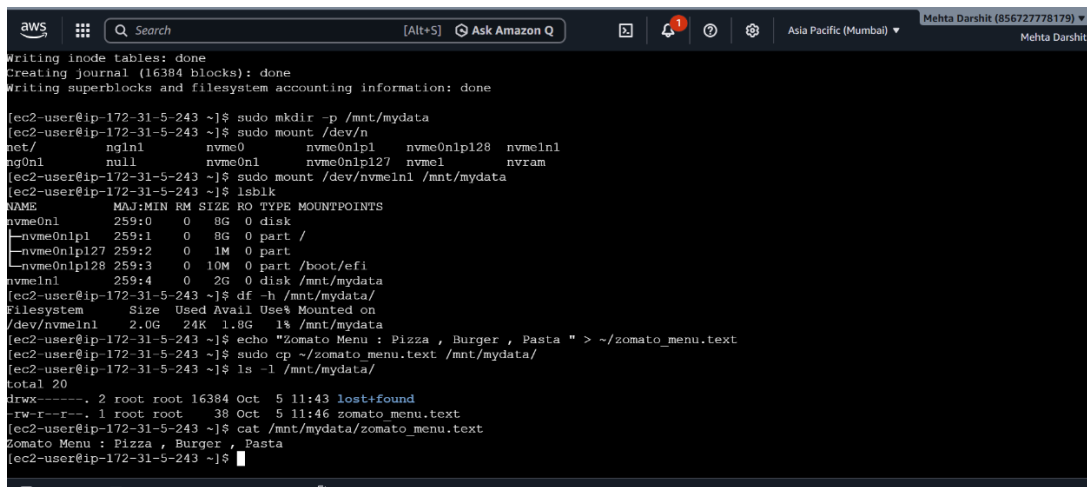
```
ls -l /mnt/mydata/
```

```
cat /mnt/mydata/zomato_menu.txt
```

Expected content:

Zomato Menu: Pizza, Burger, Biryani, Pasta

The file is now stored on your attached EBS volume rather than just in your instance's root filesystem.



```
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-172-31-5-243 ~]$ sudo mkdir -p /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ sudo mount /dev/nvme0n1p1 /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ sudo mount /dev/nvme1n1 /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ lsblk
NAME        MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1      259:0    0  8G  0 disk 
├─nvme0n1p1  259:1    0  8G  0 part /
├─nvme0n1p2  259:2    0  1M  0 part 
├─nvme0n1p3  259:3    0  10M 0 part /boot/efi
└─nvme0n1p4  259:4    0  2G  0 disk /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ df -h /mnt/mydata/
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme1n1    2.0G  24K  1.8G   1% /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$ echo "Zomato Menu : Pizza , Burger , Pasta " > ~/zomato_menu.txt
[ec2-user@ip-172-31-5-243 ~]$ sudo cp ~/zomato_menu.txt /mnt/mydata/
[ec2-user@ip-172-31-5-243 ~]$ ls -l /mnt/mydata/
total 20
drwx-----, 2 root root 16384 Oct  5 11:43 lost+found
-rw-r--r--, 1 root root   38 Oct  5 11:46 zomato_menu.txt
[ec2-user@ip-172-31-5-243 ~]$ cat /mnt/mydata/zomato_menu.txt
Zomato Menu : Pizza , Burger , Pasta
[ec2-user@ip-172-31-5-243 ~]$
```

**Tasks-5.**Detach the EBS volume from your EC2 instance using the AWS Console, then reattach it to the same or a different instance. Describe what happens to the data you stored and why this behavior is useful for apps like Flipkart or Myntra order tracking.

### A. Prepare the volume

Before detaching, unmount it from the instance:

```
sudo umount /mnt/mydata
```

In the AWS Console:

1. Open EC2 → Volumes.
2. Select your EBS volume.
3. Choose Actions → Detach volume.

4. Wait until its state becomes Available.

## B. Reattach the volume

1. Select the volume.
2. Choose Actions → Attach volume.
3. Select the same EC2 instance (or another compatible instance in the same Availability Zone).
4. Attach the volume.

Reconnect to the instance and identify its device name:

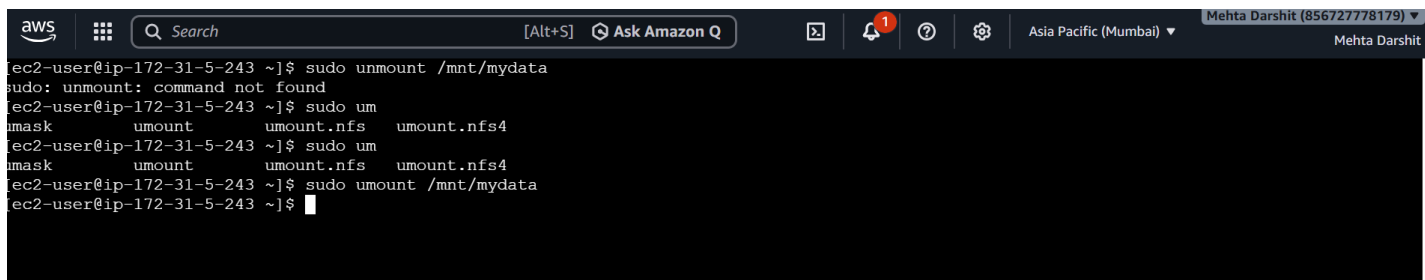
lsblk

Mount the existing filesystem again, using the correct device name:

```
sudo mount /dev/nvme1n1 /mnt/mydata
```

Check that your file is still there:

```
cat /mnt/mydata/zomato_menu.txt
```



```
aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Mehta Darshit (856727778179) Mehta Darshit
[ec2-user@ip-172-31-5-243 ~]$ sudo umount /mnt/mydata
sudo: umount: command not found
[ec2-user@ip-172-31-5-243 ~]$ sudo um
umask      umount      umount.nfs      umount.nfs4
[ec2-user@ip-172-31-5-243 ~]$ sudo um
umask      umount      umount.nfs      umount.nfs4
[ec2-user@ip-172-31-5-243 ~]$ sudo umount /mnt/mydata
[ec2-user@ip-172-31-5-243 ~]$
```

```
[ec2-user@ip-172-31-5-243 ~]$ sudo mount /dev/nvme1n1 /mnt/mydata/
[ec2-user@ip-172-31-5-243 ~]$ cat /mnt/mydata/
lost+found/      zomato_menu.txt
[ec2-user@ip-172-31-5-243 ~]$ cat /mnt/mydata/zomato_menu.txt
Zomato Menu : Pizza , Burger , Pasta
[ec2-user@ip-172-31-5-243 ~]$
```